

Forecasting Intraday Contact Center Demand

A Three-Stage Forecasting Pipeline

Team 15

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Why Interval-Level Forecasting Matters

- Over 4 million calls handled per month across Synchrony's contact centers
- Dozens of contractual service-level commitments tied to call center performance
- Staffing decisions are made at the 30-minute interval level — daily averages are not actionable
- Forecast errors in either direction carry real costs: under-staffing violates SLAs; over-staffing wastes budget

Staffing to daily averages means you're systematically over-staffed in slow windows and under-staffed during the peaks that actually matter.

4M+

Calls handled per month

~23,800

Individual predictions required for August

48

Half-hour intervals per day

Two Problems, Three Stages

Most forecasting approaches treat interval-level volume as a single prediction task. We split it — because each question is driven by completely different signals.

STAGE 1

How big is the day?

Daily Anchors



STAGE 2

When in the day does activity happen?

Interval Shape

STAGE 3 combines both outputs:

Daily Total

×

Interval Share

=

Interval Forecast

Separate client models: each portfolio trained independently — different intraday shapes, different magnitudes, different CCT curves.

Three Data Sources, One Main Constraint

Source	Granularity	Time Span	Key Fields
Daily Sheets (A–D)	1 row per day per portfolio	Jan 2024 – Aug 2025	CV, CCT, SL, Abandon Rate
Interval Sheets (A–D)	1 row per 30-min interval	Apr – Jun 2025 only	CV, CCT, ABD per interval
Daily Staffing	1 row per day	Jan – Dec 2025	Agent count per portfolio

Main Issues Found in Data Audit

- Interval totals didn't reconcile with daily totals — traced to systematic missingness, especially for CCT and ABD
- 2,000+ interval rows with at least one missing field

Key Constraint

- Only 3 months of interval data (Apr–Jun 2025) — creating an out-of-distribution risk when forecasting August.

Building the August Daily Anchor

August 2025 daily totals are used directly from the provided dataset

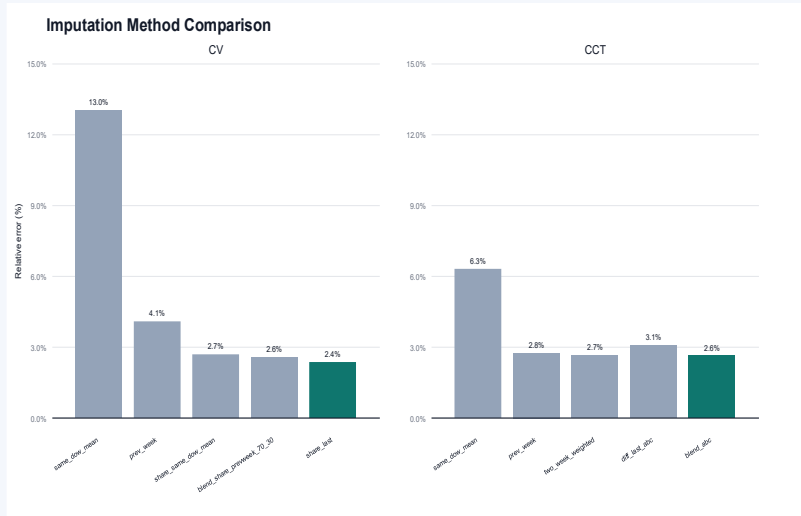
Handling Missingness: 5 Missing Rows in Portfolio D

- **Call Volume (CV)** — Scaled Portfolio D using how A+B+C moved from the same weekday one week prior.
- Tested 5 imputation methods; chose best on Aug 15–26 holdout.

CV Holdout Error: ~2.4%

- **CCT** — Prior-week Portfolio D CCT adjusted by A/B/C movement.
- Same 5-method holdout validation.

CCT Holdout Error: ~2.6%

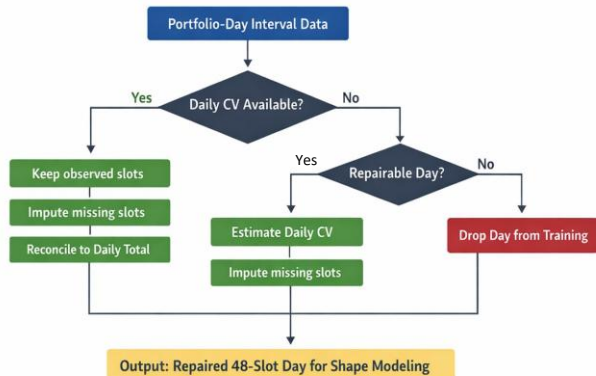


Why impute rather than drop? The daily anchor is a key feature in Stage 2 — gaps would create forecast blind spots on specific days.

Repairing the Training Data

Call Volume

- Systematic DOW × slot missingness — not random
- Interval totals ≠ daily totals (traced to missing slots)
- Multiple missingness types → custom repair algorithm



CCT & ABD

- Higher missingness than CV
- Small-denominator instability in ABD rate calculation
- Did not reconcile to daily totals — structural mismatch

Solution:

1

Clean interval structure

2

Remove unreliable observations

3

Built robust intraday patterns (using medians)

4

Smooth profiles using adjacent slots

CV: DOW × slot-index medians **CCT & ABD:** Discard raw interval magnitude — rebuild anchored to daily sheet.

Feature Selection

Goal: predict each 30-minute slot's share of the day's total call volume

Where we are in the day

slot (interval)

30-min position (0–47)

is_peak

High-demand flag

slot_sin / slot_cos

Cyclic slot encoding

Where we are in the week

dow

Day of week

dow_sin / dow_cos

Cyclic DOW encoding

How large the day is

daily_cv

Daily anchor

Cyclic encodings: sin/cos transforms make Sunday–Monday adjacent — a raw integer treats them as maximally distant.

Algorithm Comparison: Intraday Call Volume

	A	B	C	D	Avg
Ridge (Linear)	161.9%	153.6%	98.3%	126.0%	135.0%
Random Forest	26.9%	19.9%	13.7%	15.3%	18.9%
ExtraTrees	26.9%	19.9%	12.6%	15.1%	18.7%
HGB	27.7%	23.7%	12.8%	16.3%	20.1%
HGB + ET (Ours)	26.5%	20.9%	12.0%	15.1%	18.6%

Train: Apr – May 2025 Test: June 2025 Target: August 2025

Key Findings

Linear Regression fails at 135% MAPE

A linear model cannot capture the intraday curve shape.

Winner: HGB + ExtraTrees Ensemble

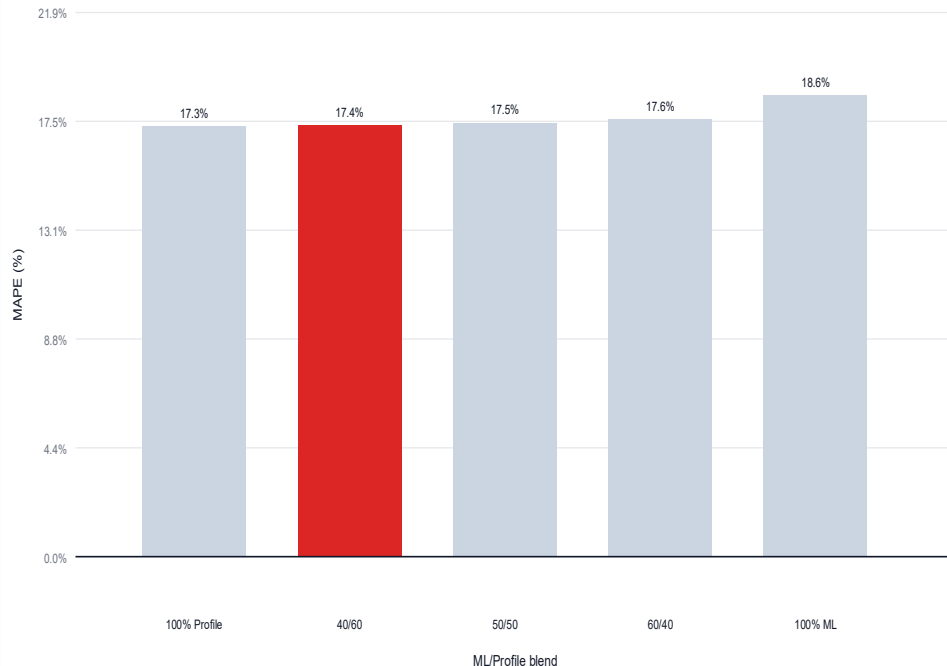
These Algorithms predict with different failure modes, averaging them led to reduced variance

Final holdout MAPE: 18.6%

Evaluated on June 2025 — the closest available analog to August given the 3-month training window.

Blending ML with Historical Profiles

CV Blend Comparison



Final Blend: 40% ML + 60% Historical Profile

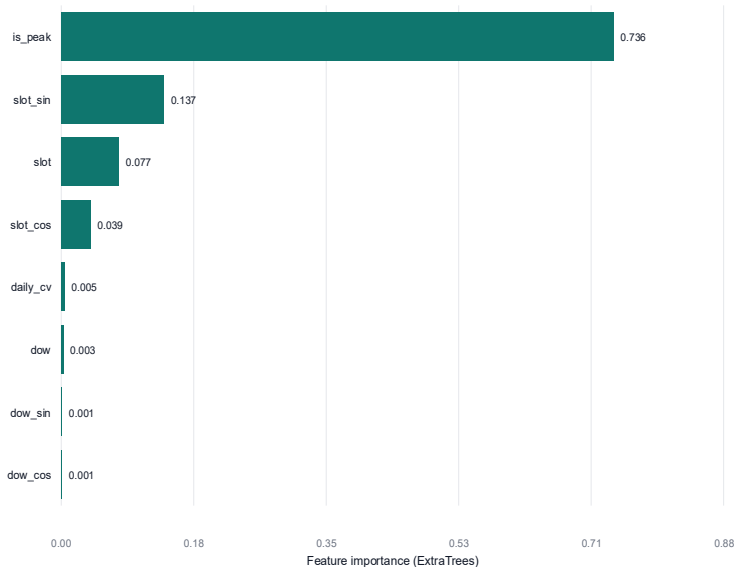
Judgement Call

Pure profiles gave almost exactly the same MAPE as 40/60 blend on June holdout – a typical month

Decision: *We chose the blend because ML produces a smoother, continuous shape estimate*

What Powers the Model

Feature Importance for CV Share Model



Goal: predict each 30-minute slot's share of the day's total call volume

Peak slot share is a core driver

Peak-hour indicator tells the model whether the interval is in the main business-day demand window

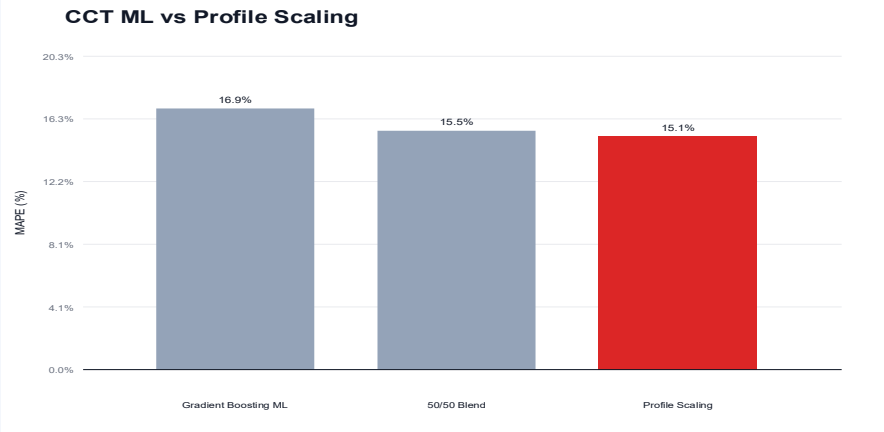
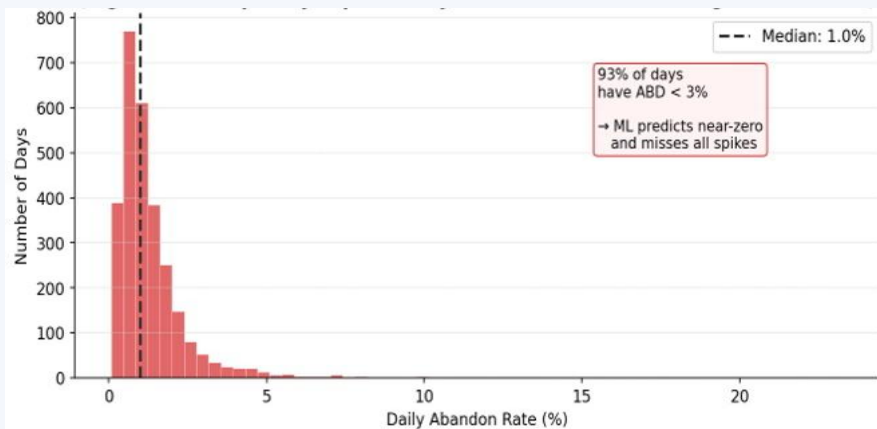
Interval position is extremely important

Interval-positioning features tells the model where each half-hour is within the day

Example

Portfolio C, Friday August 1, 2025 — peak predicted at the 2:30 PM slot. An operations manager can see exactly which features drove the shape.

CCT & ABD: When Simpler Wins



Why Profile-Based for ABD?

93% of days are under 3% abandon rate. ML models predict near-zero every time, missing spikes.

How we forecasted — Allocated daily abandoned calls with a historical intraday share pattern, then derived interval abandon rate from interval calls.

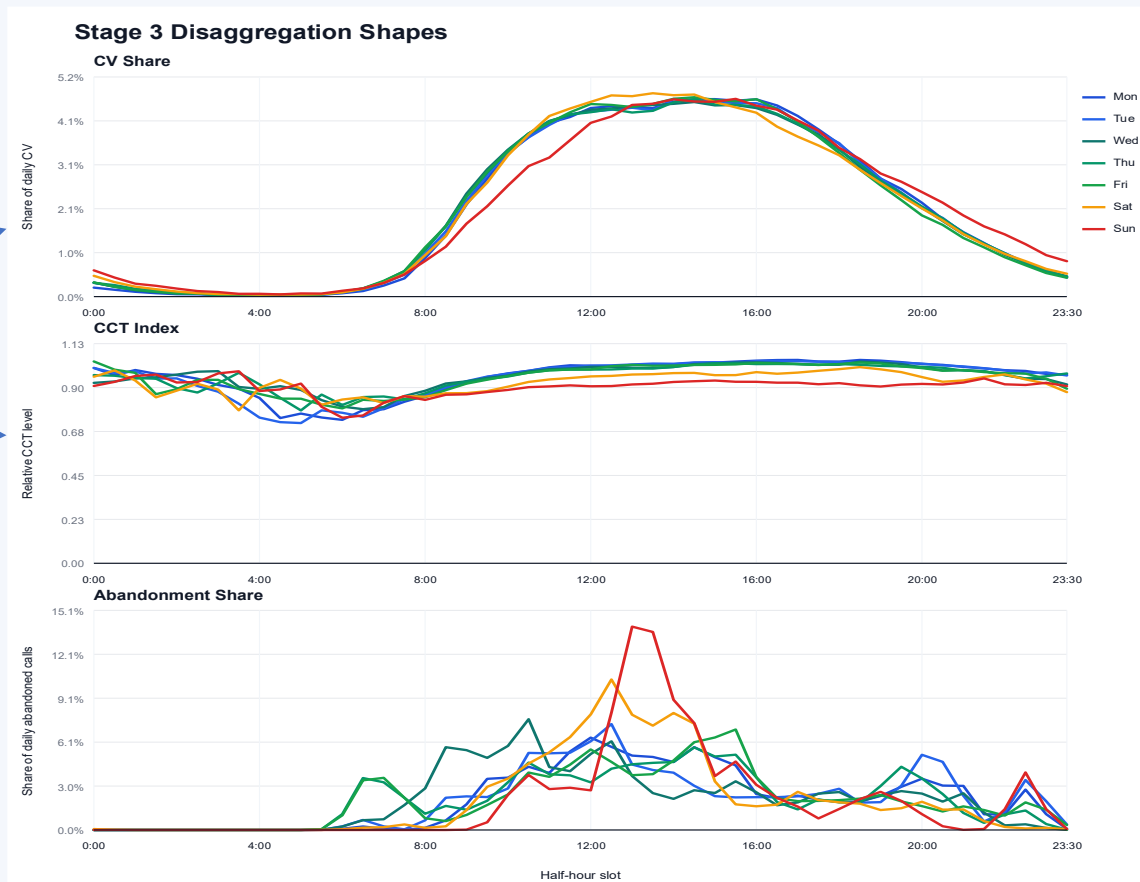
Why Profile-Based for CCT?

At the 30-minute interval level, CCT is noisy and sensitive to low call volumes. ML adds more variance.

How we forecasted — Scaled a historical intraday CCT pattern to match each day's CCT anchor.

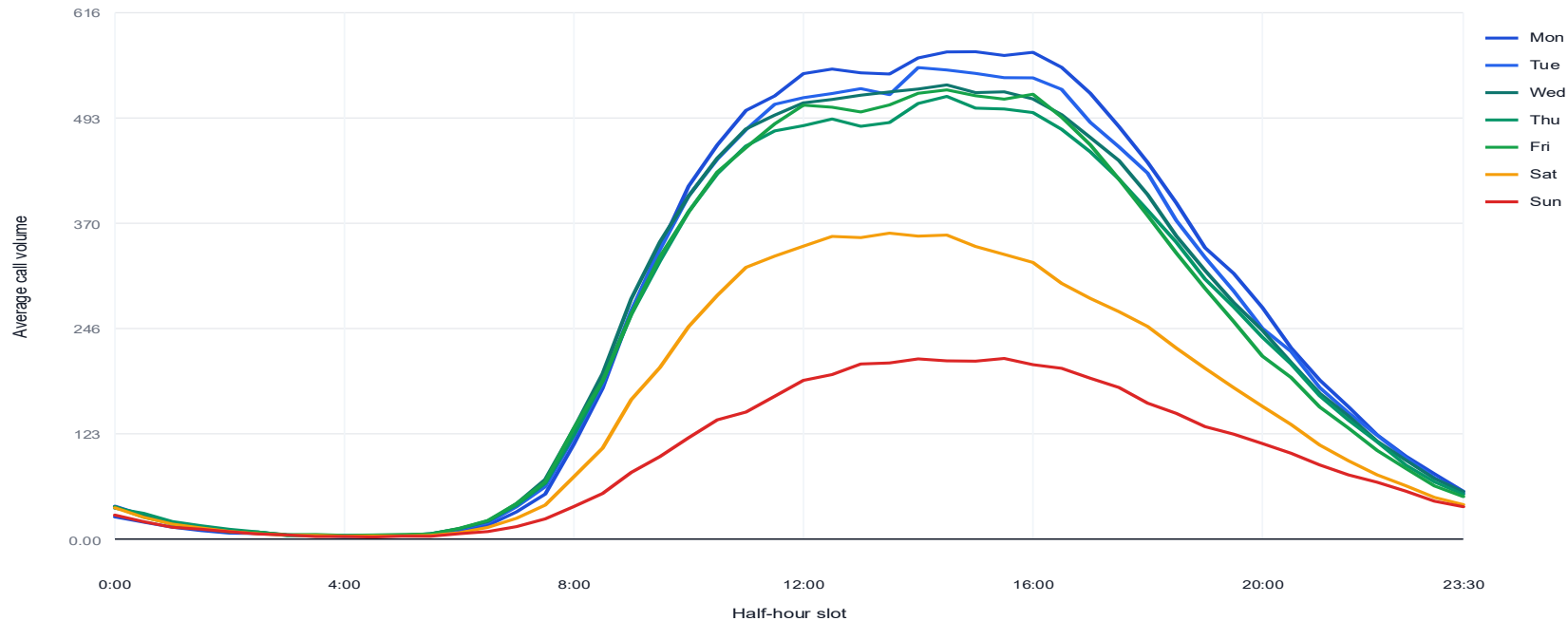
Disaggregation into 48 intervals

Daily
Anchors



Final Forecasts

Intraday CV Shape by Day of Week



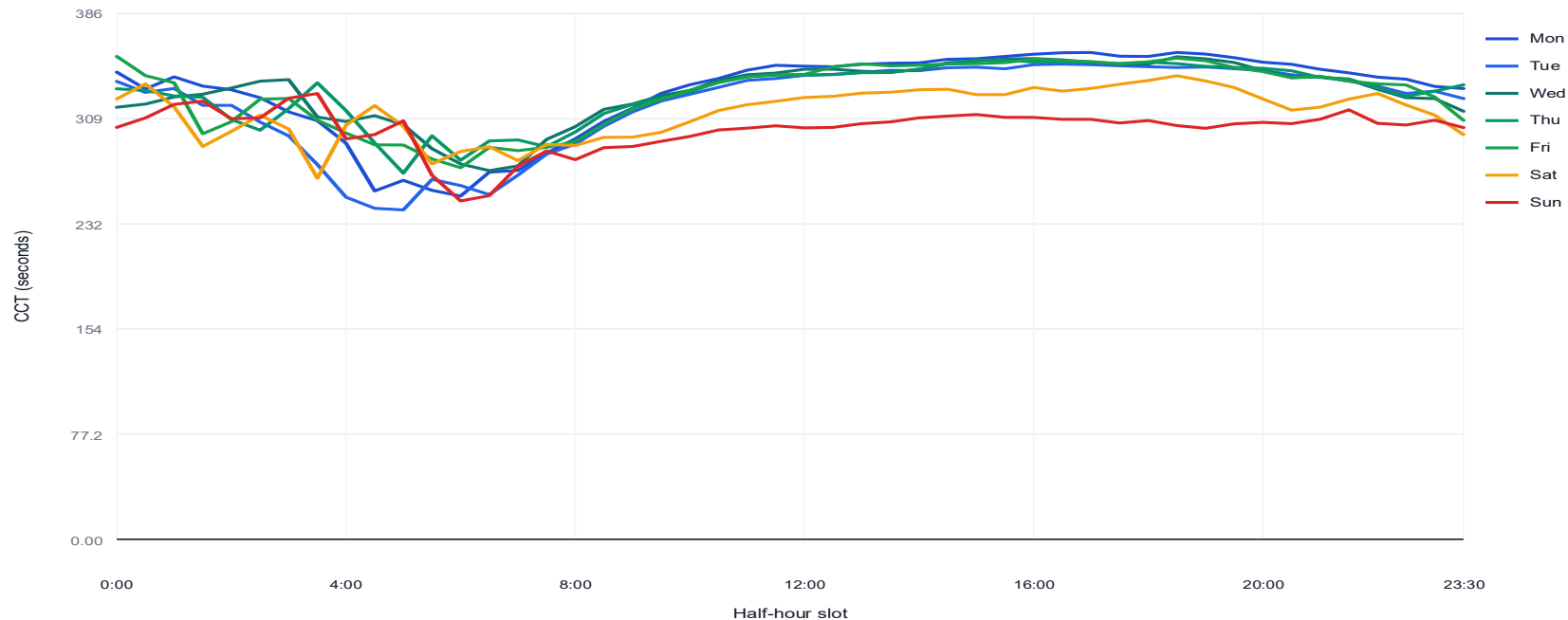
Peak around 2:30 PM daily

Monday highest volume

Weekends ~40% lower

Final Forecasts

Forecasted Interval CCT by Day of Week



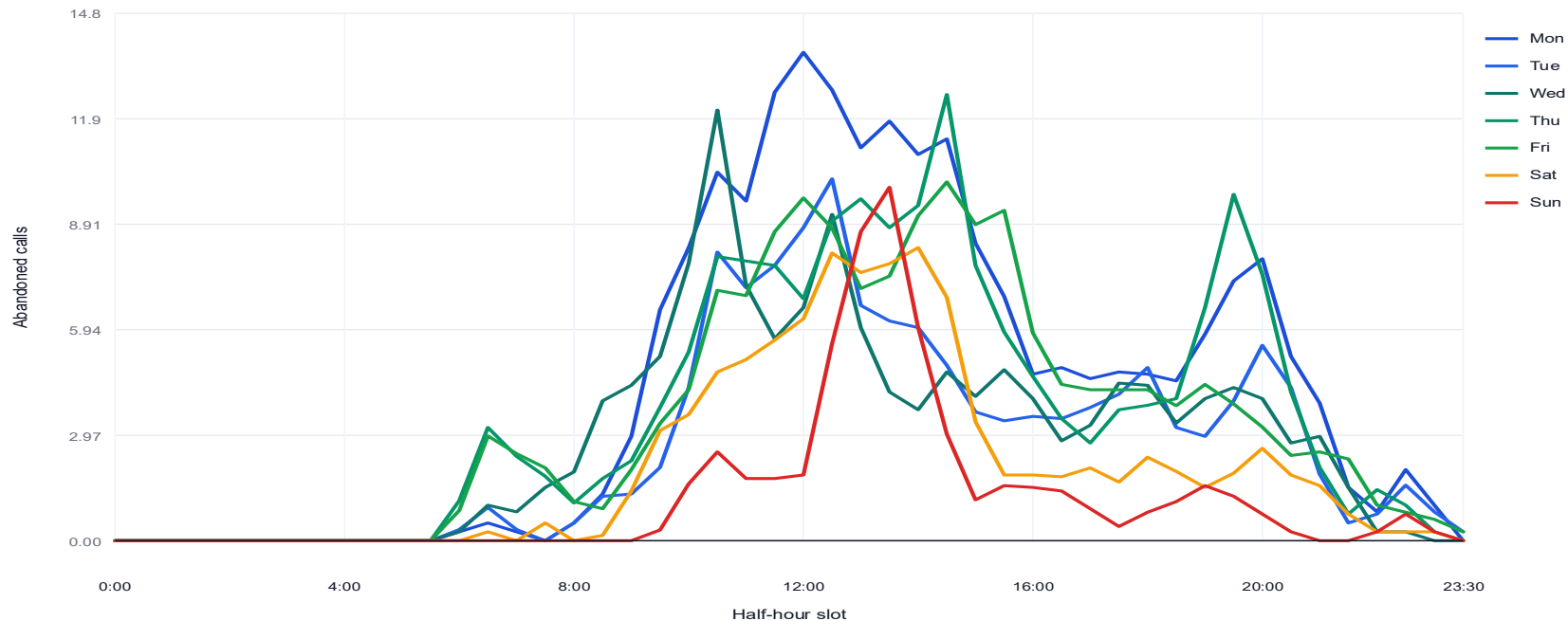
Late-afternoon CCT pressure

Consistent after 12pm

Weekends slightly lower

Final Forecasts

Forecasted Abandoned Calls by Day of Week



Peaks around midday

Monday - highest abandonment

Weekends roughly half of weekdays

August 2025: Full Picture

	Portfolio A	Portfolio B	Portfolio C	Portfolio D
Total Calls	110,613	261,572	567,384	287,700
Total Abandoned Calls	1,710	6,027	10,602	5,295
Avg Daily CCT	321s	339s	338s	335s
Avg Busiest Time	2:30 PM	2:30 PM	2:00 PM	2:30 PM
Avg Slowest Time	4:00 AM	5:00 AM	4:00 AM	4:00 AM

1.23M

Total August Calls Forecast

August 4th

Busiest Day of the Month

23.6K

Abandoned Calls

What This Means For Operations

What We Learned

- Build in two stages

- Predict **slot share** when daily totals are available in data

- For noisy metrics like CCT and ABD, historical profiles outperform ML at the interval level

- Train separately per portfolio

Why it Scales

- Modular Design

- Swap or retrain any component without rebuilding the entire pipeline

- Adding a new portfolio only requires new data, no structural changes

- When longer interval history becomes available, plug it in easily

Most importantly: Apply a slight upward bias to CV forecasts so the pipeline errs toward adequate staff coverage.

Ready to Scale

Production Feasibility: Pipeline runs end-to-end in minutes. Data dependencies are limited to daily operational totals. No custom infrastructure required.

01 Adaptive Blending

Auto-reweight the ML-profile blend each month as new data arrives. As Synchrony's portfolio mix and call patterns shift, the model stays calibrated without manual intervention.

Near-term

02 Platform Expansion

The same architecture scales directly to Health & Wellness, Lifestyle, and Home & Auto. Each new service group requires only data — no architectural changes.

Near-term

03 Seasonal Calibration

Twelve months of interval data would let the model learn summer vs. winter demand directly in training. Three months is the single biggest constraint on forecast confidence

Data availability

Thank You

We're happy to take questions.

Team 015 · Illinois Datathon 2026

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Code & Data

GitHub Repository: [DarylOkeke/datathon-final-submission](https://github.com/DarylOkeke/datathon-final-submission)

Final submission code, Jupyter notebooks, and README with full setup instructions and commands to reproduce results

Dataset (Box): <https://uofi.app.box.com/folder/368807867391>

Training data, model output template, and team ID assignments